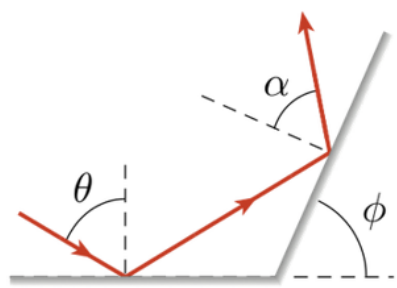
Module 33

Question 1

A beam of light is incident on a mirror lying flat on the ground at an angle of $\theta = 74°$ with respect to the normal. A second mirror is aligned with one edge of the first mirror and tilted up at an angle of $\phi = 40°$. At what angle α with respect to the normal does the ray reflect off the second mirror?



The problem involves two reflections.

1. The angle of incidence is $\theta = 74°$ with respect to the normal. Since the angle of reflection is equal to the angle of incidence, the light will reflect off the first mirror at an angle of 74° with respect to the normal.

2. This reflected ray now acts as the incident ray for the second mirror. The second mirror is tilted at an angle $\phi = 40°$ from the horizontal. This means the normal to the second mirror is also tilted 40° from the vertical.

3. To find the angle of incidence α' for the second mirror, you need to consider the difference between the first reflection angle 74° and the tilt of the second mirror 40°. Because the two mirrors share one edge, the tilt of the second mirror changes the normal's angle with respect to the first reflection.

The light ray's path and the normal to the second mirror create a supplementary pair. Given that the sum of angles in a supplementary pair is 180°, the angle between the path of the reflected ray from the first mirror (which is 74° with respect to the normal of the first mirror) and the normal to the second mirror is $180° - 74° = 106°$.

Now, since the normal to the second mirror is tilted 40° from the vertical, we need to find the actual angle of incidence α' to the second mirror. The angle α' will be the difference between 106° and the tilt angle 40°, which is $106° - 40° = 66°$.

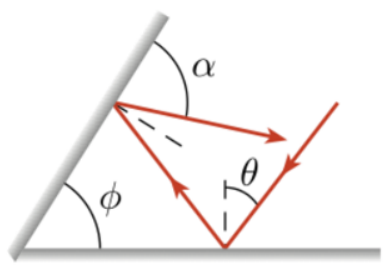
This is the angle of incidence with respect to the normal of the second mirror, and by the law of reflection, the angle of reflection α will be the same with respect to the normal of the second mirror:

$$\alpha = \alpha' = 66^\circ$$

So the ray will reflect off the second mirror at an angle of 66° with respect to the normal of the second mirror.

Question 2

The figure (not to scale) shows two plane mirrors and an incident light ray. Find the angle α if $\phi = 58^\circ$ and $\theta = 29^\circ$.



According to the law of reflection:

$$\varphi = \varphi' \ (1)$$

φ and α are related by:

$$\varphi = 90^\circ - \alpha \ (2)$$

We can use (1) and (2) to establish a relationship between β, φ, and α:

$$2 \times \varphi + \alpha + \beta = 2 \times (90^\circ - \alpha) + \beta = 180^\circ$$

which will lead to:

$$\beta = \alpha$$

If we proceed the same way, we can prove that:

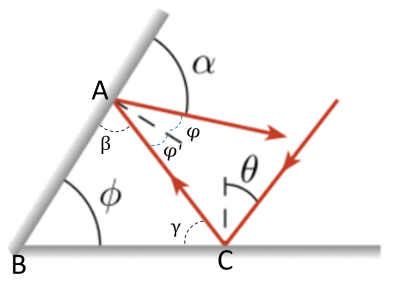
$$\gamma = 90^\circ - \theta$$

Grouping all these equations, we can write a relationship between the different angles in triangle ABC:

$$\gamma + \beta + \phi = 180^\circ$$

We can easily show that:

$$\alpha = 61^\circ$$



Question 3

Suppose the index of refraction of a certain medium is 1.87. Calculate the speed of light in that medium and supply the numerical prefix missing from the blank below.

$$v = ____ \times 10^8 \text{ m/s}$$

To calculate the speed of light in the given medium, we can use the formula:

$$v = \frac{c}{n}$$

Where:

- v is the speed of light in the medium,
- c is the speed of light in vacuum (approximately 3.00×10^8 m/s),
- n is the index of refraction of the medium.

Substituting the given values:

$$v = \frac{3.00 \times 10^8 \text{ m/s}}{1.87}$$

$$v \approx 1.60 \times 10^8 \text{ m/s}$$

Question 4

Light with a wavelength of 507 nm in vacuum enters a material with an index of refraction of 2.5. Calculate its wavelength in nm in the medium.

$$\lambda = ____ \text{ nm}$$

When light enters a medium with an index of refraction n, its wavelength in the medium is given by:

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}$$

Where:

- λ_{medium} is the wavelength of light in the medium.
- λ_{vacuum} is the wavelength of light in vacuum.
- n is the index of refraction of the medium.

Given:

- $\lambda_{\text{vacuum}} = 507\,\text{nm}$
- $n = 2.5$

Substituting the given values:

$$\lambda_{\text{medium}} = \frac{507\,\text{nm}}{2.5} = 202.8\,\text{nm}$$

Question 5

A ray of light traveling in a fluid with index of refraction $n_1 = 1.3$ is incident at an angle of $33°$ with respect to the normal on a transparent material with index of refraction $n_2 = 2.6$. Calculate the refraction angle in degrees.

$$\theta_2 = \underline{\quad\quad}^\circ$$

To calculate the refraction angle (θ_2), we can use Snell's law, which states:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

Where:

- n_1 and n_2 are the indices of refraction of the two media,
- θ_1 is the angle of incidence,
- θ_2 is the angle of refraction.

Given:

- $n_1 = 1.3$,
- $\theta_1 = 33°$,
- $n_2 = 2.6$.

We can rearrange Snell's law to solve for θ_2:

$$\theta_2 = \sin^{-1}\left(\frac{n_1}{n_2}\sin(\theta_1)\right)$$

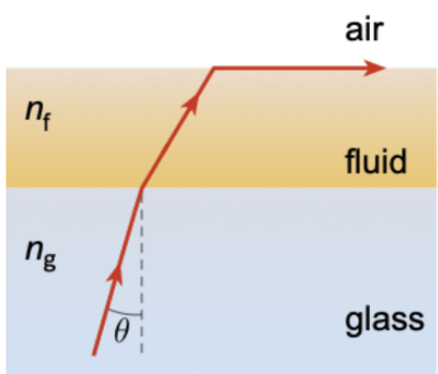
Substituting the given values:

$$\theta_2 = \sin^{-1}\left(\frac{1.3}{2.6} \times \sin(33^\circ)\right) \approx 16.14^\circ$$

Question 6

The figure shows a light ray in a glass slab. The ray passes into a fluid layer and then undergoes total internal reflection at the fluid-air boundary. Calculate the angle θ in degrees if $n_g = 2.77$ and $n_f = 1.74$.

[Hint: Work backwards.]



To solve for the angle θ in the glass slab, we first need to find the critical angle θ_c at the fluid-air boundary, which is given by:

$$\theta_c = \arcsin\left(\frac{1}{n_f}\right)$$

Given $n_f = 1.74$, we can calculate θ_c:

$$\theta_c = \arcsin\left(\frac{1}{1.74}\right)$$

$$\theta_c \approx \arcsin(0.5747)$$

$$\theta_c \approx 35.61^\circ$$

Now, since the light undergoes total internal reflection at the fluid-air boundary, the angle of incidence θ_f within the fluid must be equal to the critical angle:

$$\theta_f = \theta_c = 35.61^\circ$$

Finally, we use Snell's Law again to find the angle θ in the glass:

$$n_g \sin(\theta) = n_f \sin(\theta_f)$$

Given $n_g = 2.77$ and $n_f = 1.74$, we can rearrange the equation to solve for θ:

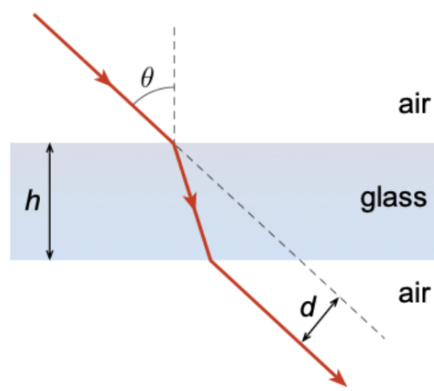
$$\sin(\theta) = \frac{n_f}{n_g} \sin(\theta_f)$$

$$\sin(\theta) = \frac{1.74}{2.77} \times \sin(35.61^\circ) \approx 21.20^\circ$$

Question 7

The figure shows a light ray incident on a pane of glass in air with $\theta = 44^\circ$. Calculate the lateral distance d in cm the ray is shifted if $n_{glass} = 6$ and $h = 9$ cm.

$d =$ ____ cm



We first use triangle ACD:

$$d = AC \times \sin(\theta - r) \ (1)$$

Now, we can turn our sight towards triangle ABC. We can easily establish that:

$$AC = \frac{h}{\cos(r)} \ (2)$$

Using (1) and (2), we can write:

$$d = \frac{h}{\cos(r)} \times \sin(\theta - r) \ (3)$$

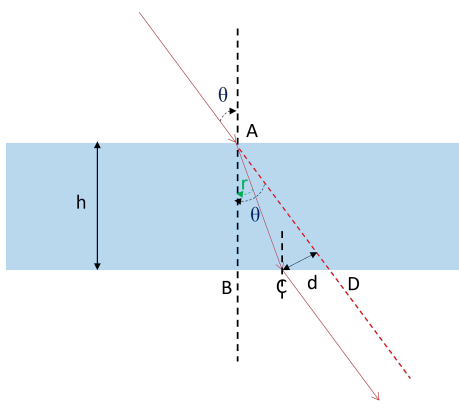
i and r are related through Snell's law by:

$$\sin(\theta) = n_{\text{glass}} \sin(r)$$

Thus,

$$r = \arcsin\left(\frac{1}{n_{\text{glass}}} \sin(\theta)\right)$$

r= 4.49 cm



Question 8

The table below shows the index of refraction for BK7 optical glass at several different wavelengths. White light is incident on the top face of a large block of this glass at an angle of 50° with respect to the normal. If the block is 0.47 m thick, by how many millimeters are beams at wavelengths 2 and 4 separated at the bottom face of the block?

Table 1: Index of refraction for BK7 optical glass at different wavelengths

Number	**Wavelength (nm)**	**Index of Refraction**
1	400	1.531
2	475	1.523
3	550	1.519
4	625	1.515

This is a dispersion problem: Its spectral component will follow its own path inside the glass slab because of dispersion (refractive index dependency

upon wavelength). Let n_1 and n_2 be the refractive indices for λ_2 and λ_4. As a consequence, the two colors will emerge at different positions on the second interface (glass-air). Let δ_1 and δ_2 be the lateral shift of color 2 and color 4. We are looking for the relative shift, which is $\Delta = \delta_2 - \delta_1$.

If we use triangles ABC and ABD, we can easily find:

$$\delta_1 = h \times \tan(r_1)$$
$$\delta_2 = h \times \tan(r_2)$$

Thus:

$$\Delta = h\left(\tan(r_2) - \tan(r_1)\right)$$

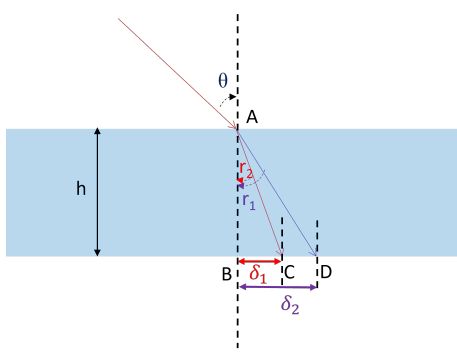
r_1 and r_2 are related to θ through Snell's law:

$$\sin(\theta) = n_1 \sin(r_1) = n_2 \sin(r_2)$$

Thus:

$$r_1 = \arcsin\left(\frac{1}{n_1}\sin(\theta)\right)$$
$$r_2 = \arcsin\left(\frac{1}{n_2}\sin(\theta)\right)$$

Using the numerical values supplied in the problem, we can easily find $\Delta = \delta_2 - \delta_1 = 1.93\,\text{mm}$.



Question 9

A ray of light is traveling in a fluid having an index of refraction 2 when it encounters a boundary with another fluid that has index of refraction 1.2. Calculate the critical angle in degrees for total internal reflection.

To calculate the critical angle for total internal reflection, we can use Snell's Law, which relates the angle of incidence (θ_c) to the indices of refraction of the two media:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

Where: - n_1 and n_2 are the indices of refraction of the two media, - θ_1 is the angle of incidence, - θ_2 is the angle of refraction.

For total internal reflection, the angle of refraction θ_2 is 90°. Therefore, $\sin(\theta_2) = 1$.

Given: - $n_1 = 2$, - $n_2 = 1.2$, - $\theta_2 = 90°$.

We can rearrange Snell's Law to solve for θ_1, the angle of incidence corresponding to total internal reflection:

$$\sin(\theta_1) = \frac{n_2}{n_1} \times \sin(\theta_2)$$

$$\sin(\theta_1) = \frac{1.2}{2} \times 1$$

$$\sin(\theta_1) = 0.6$$

Now, to find θ_1, we take the inverse sine (arcsine) of 0.6:

$$\theta_1 = \arcsin(0.6)$$

Calculating the arcsine gives us:

$$\theta_1 \approx 36.87°$$

So, the critical angle for total internal reflection is approximately 36.87°.

Question 10

A liquid-air boundary has a critical angle of 40.1° for total internal reflection. If a ray of light traveling in air has an incident angle of 29° at the liquid-air boundary, calculate the angle between the refracted ray and the surface normal.

Let θ_i be the angle of incidence and θ_r be the refraction angle, both measured with respect to the normal. From Snell's law:

$$\sin(\theta_i) = n_{\text{fluid}} \sin(\theta_r)$$

Thus:

$$\theta_r = \arcsin\left(\frac{1}{n_{\text{fluid}}} \sin(\theta_i)\right)$$

To find n_{fluid}, we will use the definition of the critical angle (θ_c):

$$\sin(\theta_c) = \frac{1}{n_{\text{fluid}}}$$

Thus:

$$\theta_r = \arcsin\left(\sin(\theta_c)\sin(\theta_i)\right) = \arcsin\left(\sin(40.1^\circ)\sin(29^\circ)\right) = 18.61^\circ$$

Question 11

Light traveling in air is incident on a fluid with index of refraction 1.95. At what angle of incidence with respect to the normal is the reflected light completely polarized? Answer in degrees.

When light is incident at a specific angle, known as the Brewster angle (θ_B), on a medium, the reflected light becomes completely polarized parallel to the surface of the medium. The Brewster angle can be calculated using the formula:

$$\tan(\theta_B) = n$$

where n is the index of refraction of the medium.

Given that the index of refraction of the fluid is $n = 1.95$, we can calculate the Brewster angle:

$$\tan(\theta_B) = 1.95$$

Taking the arctangent of both sides to solve for θ_B:

$$\theta_B = \arctan(1.95)$$

Calculating the arctangent gives us:

$$\theta_B \approx 63.6^\circ$$

So, the angle of incidence at which the reflected light is completely polarized is approximately 63.6°.

Question 12

The larger index of refraction of a given material, the `[ ]` the speed of light in the material. Further, for a given nonzero angle with respect to the normal, a larger index of refraction means the ray will bend `[ ]` from its original trajectory upon entering the material from air.

The larger index of refraction of a given material, the **lower** the speed of light in the material. Further, for a given nonzero angle with respect to the normal, a larger index of refraction means the ray will bend **more** from its original trajectory upon entering the material from air.

Question 13

When a ray of unpolarized light encounters a polarization filter, the ray's intensity is reduced by `[ ]`. If it then encounters a second polarization filter with axis perpendicular to the first filter, the intensity is `[ ]`.

When a ray of unpolarized light encounters a polarization filter, the ray's intensity is reduced by `[1/2]`. If it then encounters a second polarization filter with axis perpendicular to the first filter, the intensity is `[0]`.

Module 34

Question 1

An object is placed 21 cm to the left of a flat mirror.

(a) What is the position of the image? (Positive on the same side as the object; negative on the opposite side. Answer with an integer.)

(b) What is the magnification of the image? (Answer with an integer.)

(c) State whether the image is real or virtual.

(d) State whether the image is upright or inverted.

For a flat mirror, the laws of reflection are quite straightforward.

(a) **Position of the image:** The image in a flat mirror appears to be the same distance behind the mirror as the object is in front of it. However, by convention, distances behind the mirror are considered negative. Since the object is 21 cm in front of the mirror, the image will be 21 cm behind the mirror. The position of the image is therefore -21 cm.

(b) **Magnification of the image:** The magnification (m) in a flat mirror is always 1 because the image is the same size as the object. The magnification is positive when the image is upright, which is the case with a flat mirror. Hence, the magnification is +1.

(c) **Real or virtual image:** The image formed by a flat mirror is always virtual. A virtual image is formed when the reflected rays appear to diverge from a point behind the mirror. Since these rays do not actually come from the image (they only appear to when traced back), the image cannot be projected onto a screen and is thus virtual.

(d) **Upright or inverted image:** The image in a flat mirror is always upright relative to the object.

So the answers are: (a) -21 cm (b) +1 (c) Virtual (d) Upright

Question 2

A spherical mirror is concave and has focal length 15.6 cm. If the object distance is 7.9 cm, calculate the image distance, s', in cm.

To find the image distance (s') for a concave spherical mirror, we can use the mirror equation:

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$$

where: - f is the focal length of the mirror, - s is the object distance, - s' is the image distance.

Given: - $f = 15.6\,\text{cm}$, - $s = 7.9\,\text{cm}$.

We can rearrange the mirror equation to solve for s':

$$\frac{1}{s'} = \frac{1}{f} - \frac{1}{s}$$

$$s' = \frac{sf}{s - f}$$

Now, let's calculate s':

$$s' = \frac{7.9 \times 15.6}{7.9 - 15.6}$$

$$s' \approx -16\,\text{cm}$$

The negative sign for s' indicates that the image is formed on the same side as the object, which is characteristic of a virtual image formed by a concave mirror. So, the image distance is approximately $-16\,\text{cm}$.

Question 3

An eggplant is 2.9 cm from the vertex of a convex spherical mirror having a radius of curvature of 14 cm. Calculate the lateral magnification.

To find the image distance (s') for a convex spherical mirror, we can use the mirror equation:

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$$

where: - f is the focal length of the mirror, - s is the object distance, - s' is the image distance. Given: - $f = R/2 = -7\,\text{cm}$, - $s = 2.9\,\text{cm}$.

We can rearrange the mirror equation to solve for s':

$$\frac{1}{s'} = \frac{1}{f} - \frac{1}{s}$$

$$s' = \frac{sf}{s - f}$$

Now, let's calculate s':

$$s' = \frac{2.9 \times (-7)}{2.9 - (-7)}$$

$$s' \approx -2.05\,\text{cm}$$

Now to find the lateral magnification m defined as $m = \frac{-s'}{s}$ If we do the calculation, we get m=0.707

Question 4

A thin diverging lens has a focal length of 9.00 cm. If the object distance is 6.0 cm, the image distance is `[S ]` cm, the magnification is `[S ]`, and the image is both `[S ]` and `[S ]`.

Let's use the same approach for the thin diverging lens problem.

Given: - Focal length $f = -9.00\,\text{cm}$, - Object distance $s = 6.0\,\text{cm}$.

We'll use the lens equation:

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$$

Rearranging the lens equation to solve for s':

$$\frac{1}{s'} = \frac{1}{f} - \frac{1}{s}$$

$$s' = \frac{sf}{s - f}$$

Now, let's calculate s':

$$s' = \frac{6.0 \times (-9.00)}{6.0 - (-9.00)}$$

$$s' = \frac{-54.00}{6.0 + 9.00}$$

$$s' = \frac{-54.00}{15.00}$$

$$s' = -3.6\,\text{cm}$$

Now, let's find the lateral magnification m defined as $m = \frac{-s'}{s}$:

$$m = \frac{-(-3.6)}{6.0}$$

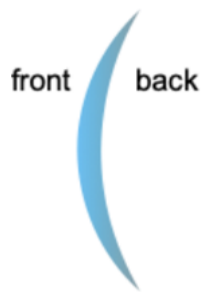
$$m = \frac{3.6}{6.0}$$

$$m = 0.6$$

So, the image distance is approximately $-3.6\,\text{cm}$, and the lateral magnification is approximately 0.6. The image is virtual (s'¡0) and upright (m¿0)

Question 5

The figure shows a contact lens made from a material with an index of refraction of 1.8. The front and back surfaces have radii of curvature of 1.8 cm and 2.6 cm, respectively. Calculate the lens' focal length in cm.



To calculate the focal length of a lens, we can use the Lensmaker's Equation, which for a lens in air is:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

where: - f is the focal length of the lens. - n is the index of refraction of the lens material. - R_1 is the radius of curvature of the lens' front surface. - R_2 is the radius of curvature of the lens' back surface.

A positive radius of curvature indicates that the center of curvature is on the same side as the outgoing light, and a negative radius of curvature indicates that the center of curvature is on the opposite side of the outgoing light.

For a contact lens, typically the front surface is convex, and the back surface is concave relative to the incoming light. Therefore, in this case, if we assume the light is coming from the front side, R_1 would be positive and R_2 would be negative.

Given: - $n = 1.8$ - $R_1 = 1.8\,\text{cm}$ (assuming it's positive) - $R_2 = 2.6\,\text{cm}$ (assuming it's negative since it's concave)

Let's plug these values into the equation and solve for f.

$$\frac{1}{f} = (1.8-1)\left(\frac{1}{1.8} - \frac{1}{2.6}\right) = 7.31\text{cm}$$

Question 6

A coin is at the bottom of a tank of fluid 148 cm deep having index of refraction 1.56. Calculate the image distance in cm as seen from directly above. [Your answer should be negative!]

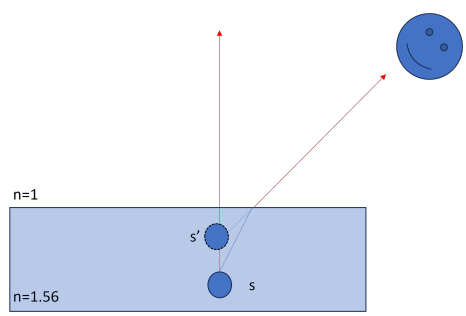
$$\frac{n_1}{s} + \frac{n_2}{s'} = \frac{(n_2 - n_1)}{R}$$

We are solving for s', here the interface is flat which means that $R = \infty$

$$\frac{n_1}{s} + \frac{n_2}{s'} = 0$$

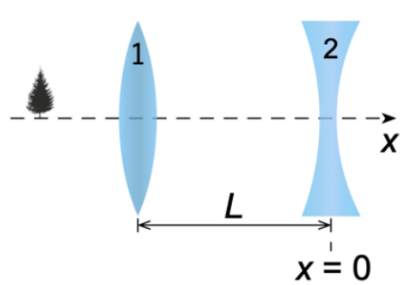
Thus,

$$s' = -\frac{s}{n_1} = -\frac{148}{1.56} = -94.87\text{cm}$$



Question 7

The figure (not to scale) shows a tiny tree 23 cm from a converging lens followed by a diverging lens with $L = 11$ cm. Find the x-position in cm of the final image if $f_1 = 9$ cm and $|f_2| = 16$ cm.



To solve for the final image position through a system of two lenses, we must first find the image position s' created by the first lens, then use that image as an object for the second lens to find the final image position.

For the first lens, which is converging, we use the lens formula:

$$\frac{1}{f_1} = \frac{1}{s_1} + \frac{1}{s_1'}$$

$$s_1' = \frac{f_1 s}{s - f_1}$$

where - f_1 is the focal length of the lens, - s_1 is the object distance from the lens, - s_1' is the image distance from the lens.

Given that the object (the tiny tree) is 23 cm from the first lens and the focal length f_1 of the first lens is 9 cm, we have:

$$s_1' = \frac{9 \times 23}{23 - 9} = 14.7857\text{cm}$$

After finding s_1', we will then find the image position created by the second lens using the image from the first lens as the object. The object distance for the second lens s_2 will be $L - s_1'$, where L is the distance between the two lenses. The focal length of the second lens f_2 is given as 16 cm, but since it is a diverging lens, it will be -16 cm in the lens formula. We will use a similar lens formula to find the final image distance s_2'.

$$s_2' = \frac{f_2(L - s_1')}{L - s_1' - f_2} = \frac{-16(11 - 14.7857)}{11 - 14.7857 - (-16)} = 4.9590\text{cm}$$

Question 8

A man has a near point of 50 cm. What power glasses in diopters should an optometrist prescribe so that he can read comfortably at a distance of 25.6 cm?

Glasses will create an image of the object located at the near point (s=25.6 cm). The image is virtual and located at s'=-50 cm (so the man can see confortably). Use the thin lens formula (To find the power in diopters (D), you convert the focal length from centimeters to meters):

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$$

$$\frac{1}{f} = \frac{1}{25.6} + \frac{1}{-50} = 0.0191 \text{ cm}^{-1} = +1.91 \text{ m}^{-1} = 1.91 \text{ dioptres}$$

Therefore, the optical power of the glasses that the doctor has to prescribe is +1.91 diopters.

Question 9

A simple magnifier has a focal length of 6.7 cm. Calculate the magnification.

$$m = \frac{25\ cm}{6.7\ cm} = 3.7313$$

Question 10

A [ ] lens corrects for near-sightedness. The image it creates is [ ] [ ]. A [ ] lens corrects for far-sightedness. The image it creates is [ ] [ ].

A **diverging** (or **concave**) lens corrects for near-sightedness. The image it creates is **virtual** and **reduced**.

A **converging** (or **convex**) lens corrects for far-sightedness. The image it creates is **virtual** and **enlarged**."

Module 36

Question 1

Monochromatic, coherent light is incident on a Young's experiment having slits separated by 0.3 mm. The screen is 2 m away and an experimenter notes that the m-th order maximum is 2.5 mm from the center line. Calculate the wavelength of the light in nanometers if $m = 3$.

In Young's double-slit experiment, the wavelength of the light can be calculated using the formula for the position of the m-th order bright fringe (constructive interference) on the screen:

$$y_m = \frac{m\lambda D}{d}$$

where - y_m is the distance from the central maximum to the m-th order bright fringe, - m is the order number of the fringe, - λ is the wavelength of the light, - D is the distance from the slits to the screen, - d is the separation between the slits.

We are given: - $y_m = 2.5$ mm, - $m = 3$, - $D = 2$ m, - $d = 0.3$ mm.

First, we need to convert y_m and d into meters to match units with D:

- $y_m = 2.5 \times 10^{-3}$ m, - $d = 0.3 \times 10^{-3}$ m.

Now, we can rearrange the formula to solve for λ:

$$\lambda = \frac{y_m d}{mD} = 125 \text{ nm}$$

Question 2

Two slits spaced 0.33 mm apart 72 cm from a screen. The slits are illuminated with coherent light of wavelength 477 nm. What is the distance (in mm) between the $m = +3$ and $m = +5$ dark lines of the interference pattern on the screen?

In a Young's double-slit experiment, the distance between the dark fringes on the interference pattern can be calculated using the formula for dark fringes:

$$y_{\text{dark}} = \left(m + \frac{1}{2}\right) \frac{\lambda D}{d}$$

where: - y_{dark} is the distance from the central maximum to the dark fringe of order m, - m is the order number of the dark fringe (half-integer values like 3.5, 4.5, etc., for dark fringes), - λ is the wavelength of the light used, - D is the distance from the slits to the screen, - d is the separation between the slits.

We need to find the distance between the $m = +3$ and $m = +5$ dark fringes.

The distance Δy between these two dark fringes will be the difference in their respective distances from the central maximum:

$$\Delta y = \left(5 + \frac{1}{2}\right) \frac{\lambda D}{d} - \left(3 + \frac{1}{2}\right) \frac{\lambda D}{d} = 2\frac{\lambda D}{d}$$

Given: - $\lambda = 477$ nm (which is 477×10^{-9} m), - $D = 72$ cm (which is 0.72 m), - $d = 0.33$ mm (which is 0.33×10^{-3} m).

We can now calculate Δy in meters and then convert it to millimeters.

$$\Delta y = 2 \times \frac{477 \times 10^{-9} \times 72 \times 10^{-2}}{0.33 \times 10^{-3}} = 2.0815 \times 10^{-3} \text{ m} = 2.0815 \text{ mm}$$

Question 3

A flat sheet of glass $n_{\text{glass}} = 1.61$ has a thin coating of material applied to its upper surface. If the refractive index of the coating is 1.27 what minimum thickness (in nm) should this coating be in order to be nonreflective to light of vacuum wavelength 557 nm?

To create a non-reflective coating on a lens, the thickness of the coating (t) should be such that it causes destructive interference for light reflected from the top and bottom surfaces of the coating. This typically requires a coating thickness of one-quarter the wavelength of the light in the coating material, known as a quarter-wave layer. The destructive interference effectively cancels out the reflected light at the specific wavelength, reducing reflection and glare.

The condition for destructive interference in a thin film is given by:

$$2nt = (m + \frac{1}{2})\lambda$$

where -m is the fringe order (m=0, in our case, minimum) - t is the thickness of the coating, - λ is the wavelength of the light in vacuum, - n is the refractive index of the coating material.

Thus:

$$t = \frac{\lambda}{4n} = \frac{557}{4 \times 1.27} = 109.6457 \text{ nm}$$

Question 4

A silicon solar cell with an index of refraction of 3.88 is coated with an anti-reflection thin film having an index of refraction of 1.55. Calculate the minimum film thickness in nm that produces the least reflection at a wavelength of 480 nm.

If we proceed the same way as in the previous question. We can use the following equation (m=0) to find t:

$$2nt = (m + \frac{1}{2})\lambda$$

$$t = \frac{\lambda}{4n} = \frac{480}{4 \times 1.55} = 77.4194 \text{ nm}$$

Question 4

Artwork in museums should be behind glass that is "anti-reflective," or "non-glare." This means coating the outer surface of the glass (index of refraction 1.5) with a thin film of refractive index 3. What is the minimum thickness (but not zero) of the film (in nm) to cancel light of wavelength 501 nm?

You might be tempted to use the same equation as in the previous two questions. Be careful here: light that gets reflected upfront (without penetrating the thin layer with $n = 3$) will incur a π phase shift (it was traveling from low n to high n). The second ray (the one that penetrated the thin layer) will strike the interface (thin layer-glass). Since the glass has a lower refractive index than the thin layer, this ray will not incur any phase shift. The condition for interference in this case will be:

$$2nt = m\lambda$$

for m=1,

$$t = \frac{\lambda}{2n} = 83.5 \text{ nm}$$

Question 5

In a double slit experiment, decreasing the distance between slits [ ] the distance between bright fringes on the screen. Decreasing the distance to the screen [ ] the distance between bright fringes on the screen.

In a double-slit experiment, decreasing the distance between slits **increases** the distance between bright fringes on the screen. Decreasing the distance to the screen **increases** the distance between bright fringes on the screen.

Question 6

A single slit experiment involves monochromatic, coherent light of wavelength 513 nm incident on an aperture of width 0.38 mm. If the screen is 2.1 m away, determine the distance in mm from the central maximum to the m =+3 dark line.

To solve the problem, let's plug the given values into the formula:

$$y_m = \frac{m\lambda D}{a}$$

Given: - $\lambda = 513$ nm (wavelength of light), - $D = 2.1$ m (distance from the slit to the screen), - $a = 0.38$ mm (width of the aperture), - $m = +3$ (order number of the dark fringe).

First, let's convert all units to meters: - $\lambda = 513 \times 10^{-9}$ m - $a = 0.38 \times 10^{-3}$ m

Now, let's calculate y_m:

$$y_m = \frac{(3)(513 \times 10^{-9})(2.1)}{0.38 \times 10^{-3}} = 2.9 \text{ mm}$$

Question 7

Monochromatic light illuminates a diffraction grating having 6,440 slits per centimeter. If the angular position of the second-order bright bands is 47°, what is the wavelength of the light (in nm)?

To find the wavelength of the light, we'll use the formula for the angular position of the m-th order bright fringe in a diffraction grating:

$$d\sin(\theta) = m\lambda$$

Given: - $d = \frac{1}{6440}$ cm (slit spacing), - $\theta = 47°$ (angular position of the second-order bright fringe), - $m = 2$ (second-order bright fringe).

First, let's convert the angular position from degrees to radians:

$$\theta_{\text{rad}} = \frac{47° \times \pi}{180}$$

Now, let's solve for λ:

$$\lambda = \frac{d\sin(\theta)}{m} = \frac{10^{-2}}{6440}\frac{\sin(47)}{2} = 3.6568 \text{ mm}$$

Question 8

The light emitted from an excited atom has two principal wavelengths: 589.21 nm and 588.54 nm. If you wish to resolve these two emission lines in first order using a diffraction grating that is 1.114 cm in length, what

minimum number of slits per centimeter must the grating have?

Given: - $m = 1$ (order number of the bright fringe), - $\Delta\lambda = \lambda_1 - \lambda_2$ (difference in wavelengths), - $R = \frac{\lambda_1}{\Delta\lambda}$ (resolving power of the grating), - diffraction grating length = 1.114 cm.

First, we need to calculate R using the given wavelengths. Then, we can find the minimum number of slits per centimeter by dividing R by the length of the diffraction grating.

Let's perform the calculations.

First, let's calculate the resolving power R using the formula:

$$R = \frac{\lambda_1}{\Delta\lambda}$$

$$R = \frac{589.21\,\text{nm}}{589.21\,\text{nm} - 588.54\,\text{nm}}$$

Now, we can calculate the minimum number of slits per centimeter:

$$\text{number of slits per cm} = \frac{R}{\text{grating length}}$$

$$R = \frac{589.21\,\text{nm}}{589.21\,\text{nm} - 588.54\,\text{nm}} = \frac{589.21\,\text{nm}}{0.67\,\text{nm}} \approx 879.17$$

Now, let's calculate the minimum number of slits per centimeter:

$$\text{number of slits per cm} = \frac{879.17}{1.114\,\text{cm}} \approx 789.68\,\text{slits/cm}$$

Therefore, the minimum number of slits per centimeter required to resolve the two emission lines is approximately 789.68 slits/cm.

Question 9

X-rays of wavelength 0.24 nm incident at a grazing angle of 16.1° on an unknown crystal produces a first-order interference fringe. Calculate the interatomic spacing between the Bragg planes in nanometers.

To calculate the interatomic spacing between the Bragg planes, we can use Bragg's law:

$$2d\sin(\theta) = m\lambda$$

where: - d is the interatomic spacing between the Bragg planes, - θ is the grazing angle of incidence, - m is the order of the fringe, - λ is the wavelength of the incident X-rays.

Given: - $\lambda = 0.24$ nm, - $\theta = 16.1°$ (grazing angle of incidence), - $m = 1$ (first-order interference fringe).

We need to solve for d. Let's rearrange Bragg's law to solve for d:

$$d = \frac{m\lambda}{2\sin(\theta)}$$

Then we can plug in the given values to find d in nanometers.

First, let's convert the grazing angle from degrees to radians:

$$\theta_{\text{rad}} = \frac{16.1^\circ \times \pi}{180}$$

Now, we can use Bragg's law to find the interatomic spacing d:

$$d = \frac{m\lambda}{2\sin(\theta_{\text{rad}})}$$

Let's perform the calculations.

Given: - Wavelength of the X-rays, $\lambda = 0.24$ nm - Grazing angle of incidence, $\theta = 16.1^\circ$ - Order of the fringe, $m = 1$

First, let's convert the grazing angle from degrees to radians:

$$\theta_{\text{rad}} = \frac{16.1^\circ \times \pi}{180}$$

Now, we can use Bragg's law to find the interatomic spacing d:

$$d = \frac{m\lambda}{2\sin(\theta_{\text{rad}})}$$

$$d \approx 0.451\,\text{nm}$$

Therefore, the interatomic spacing between the Bragg planes is approximately 0.451 nm.

Question 10

Your pupil has a diameter of 5 mm. Using 600 nm light, what is the resolving power of your eye? Express your answer in milli-degrees.

Given: - $D = 5$ mm $= 5 \times 10^{-3}$ m - $\lambda = 600$ nm $= 600 \times 10^{-9}$ m

Using the formula:

$$\text{Resolving power} = \frac{1.22 \times \lambda}{D}$$

Now, let's perform the calculations with the given adjustments.

$$D = 5 \times 10^{-3}\,\text{m}$$

$$\lambda = 600 \times 10^{-9}\,\text{m}$$

$$\text{Resolving power} = 146.4 \times 10^{-6}\,\text{radians}$$

Now, let's convert the resolving power from radians to milli-degrees using the conversion factor:

$$\text{Resolving power in milli-degrees} = \text{Resolving power in radians} \times \frac{180}{\pi} \times 1000$$

$$\text{Resolving power in milli-degrees} \approx 8.41\,\text{milli-degrees}$$

Question 10

What aperture telescope would be required to just barely resolve two astronauts standing 2.4 m apart on the moon from a space station orbiting the moon at a distance 100 km above them? Assume light of wavelength 525 nm. Answer in centimeters.

To resolve two astronauts standing 2.4 m apart on the moon from a space station orbiting the moon at a distance 100 km above them, we can use the Rayleigh criterion for resolving power:

$$\text{Resolving power} = \frac{1.22\lambda}{D}$$

where: - λ is the wavelength of light, - D is the diameter of the aperture of the telescope.

Given: - $\lambda = 525$ nm, - distance from the space station to the astronauts $(L) = 100$ km $= 100 \times 10^3$ m, - distance between the astronauts $(d) = 2.4$ m.

The aperture telescope required to just barely resolve the two astronauts corresponds to the resolving power equal to the angular separation between the astronauts, given by:

$$\theta = \tan^{-1}\left(\frac{d}{L}\right)$$

Now, let's solve for D.

First, let's calculate the angular separation θ between the astronauts using the formula:

$$\theta = \tan^{-1}\left(\frac{d}{L}\right)$$

where: - $d = 2.4$ m (distance between the astronauts), - $L = 100 \times 10^3$ m (distance from the space station to the astronauts).

Let's calculate θ.

Given: - $d = 2.4$ m (distance between the astronauts), - $L = 100 \times 10^3$ m (distance from the space station to the astronauts).

Using the formula:

$$\theta = \tan^{-1}\left(\frac{d}{L}\right)$$

$$\theta = \tan^{-1}\left(\frac{2.4}{100 \times 10^3}\right)$$

Now, let's calculate θ.

$$\theta = \tan^{-1}\left(\frac{2.4}{100 \times 10^3}\right)$$

$$\theta \approx \tan^{-1}\left(2.4 \times 10^{-5}\right)$$

Now, let's calculate θ.

$$\theta \approx \tan^{-1}\left(2.4 \times 10^{-5}\right)$$

$$\theta \approx 2.4 \times 10^{-5}$$

Now, let's calculate the resolving power using the Rayleigh criterion formula:

$$\text{Resolving power} = \frac{1.22\lambda}{D}$$

We want the resolving power to be equal to θ, so we can equate the two:

$$2.4 \times 10^{-5} = \frac{1.22 \times 525 \times 10^{-9}}{D}$$

Now, we can solve for D.

$$\text{Resolving power} = \frac{1.22\lambda}{D}$$

We want the resolving power to be equal to θ, so we can equate the two:

$$2.4 \times 10^{-5} = \frac{1.22 \times 525 \times 10^{-9}}{D}$$

Now, let's solve for D.

$$D = \frac{1.22 \times 525 \times 10^{-9}}{2.4 \times 10^{-5}}$$

$$D = \frac{1.22 \times 525}{2.4} \times 10^{-4}$$

$$D \approx 0.266\,\text{m}$$

Now, let's convert the diameter from meters to centimeters.

$$D \approx 0.266\,\text{m}$$

$$D \approx 26.6\,\text{cm}$$

Therefore, the aperture telescope required to just barely resolve the two astronauts standing 2.4 m apart on the moon from a space station orbiting the moon at a distance 100 km above them, using light of wavelength 525 nm, is approximately 26.6 cm.

Question 11

As the number of lines per centimeter increases, the interference fringes created by a diffraction grating [ ]. As the distance between the grating and the screen increases, the interference fringes [ ].

As the number of lines per centimeter increases, the interference fringes created by a diffraction grating **move farther apart** . As the distance between the grating and the screen increases, the interference fringes **move farther apart** .